

MATHS OF CRYPTOGRAPHY

What is Cryptography? Cryptography is the mathematical science of protecting information by transforming readable text into secret codes, ensuring that only authorized recipients can read it. From ancient secret messages to modern digital security, cryptography keeps data safe!

1. Fundamental Terms in Cryptography

Term	Definition	Example
Plaintext	The original, readable message before encryption.	<i>CAT</i> or <i>HELLO PLANRS</i>
Encryption	The mathematical process of converting Plaintext into Ciphertext (<i>Plaintext</i> → <i>Ciphertext</i>).	Shifting letters forward by a set key.
Ciphertext	The encoded, scrambled message that appears unreadable.	<i>FDW</i> or <i>KHOOR SODQUV</i>
Decryption	The reverse process of converting Ciphertext back to readable Plaintext (<i>Ciphertext</i> → <i>Plaintext</i>).	Shifting letters backward using the secret key.

2. Caesar Cipher: Shift the Alphabet!

The **Caesar Cipher** is one of the oldest symmetric encryption techniques, named after Julius Caesar. It works by shifting every letter in the plaintext forward or backward by a fixed number of positions along the alphabet.

Example: Shift by +3

Let's encrypt the word **CAT** using a shift key of $k = 3$:

- **C** shifts 3 steps forward (*C* → *D* → *E* → *F*) → **F**
- **A** shifts 3 steps forward (*A* → *B* → *C* → *D*) → **D**
- **T** shifts 3 steps forward (*T* → *U* → *V* → *W*) → **W**

Result: *CAT* → *FDW*

3. Modular Arithmetic: Wrap-Around Like a Clock!

What happens when shifting a letter goes past **Z**? In cryptography, we use **modular arithmetic** (clock arithmetic):

- Just like a 12-hour clock wraps around to **1** after 12 (rather than 13), the 26-letter alphabet wraps around to **A** after **Z**!
- *Example:* Shifting **Z** by +1 gives **A** (*Z* → *A*). Shifting **Y** by +3 gives **B** (*Y* → *Z* → *A* → *B*).
- Mathematically, encryption with key k is expressed as: $C = (P + k) \bmod 26$.

4. Why Caesar's Secret is Not So Secret!

Since there are only 26 letters in the English alphabet, there are only **25 possible non-zero shifts!**

- An eavesdropper or spy can easily break a Caesar cipher by testing all 25 possible shifts in a matter of seconds (a *brute-force attack*).
- Because of this weakness, modern communication requires much more advanced mathematical algorithms!

5. Preview of Modern Cryptography: RSA & The Magic Padlock

Modern encryption algorithms like **RSA** solve the security flaws of simple shift ciphers using **Asymmetric (Public-Key) Cryptography**:

Public Key (The Open Padlock)	Private Key (The Secret Key)					
Shared publicly with anyone who wants to send you a message.	Used exclusively to lock/encrypt messages.	Anyone can lock the padlock, but nobody can open it with the public key alone!	Kept strictly secret by the owner.	Used exclusively to unlock/decrypt messages.		Only the person holding the private key can unlock and read the received message!

Cadet Challenge: Decode the Secret Message!

Apply your decoding skills to decrypt the ciphertext: **KHOOR SODQUV** using a backward shift of **k = 3**:

- **K (-3) → H, H (-3) → E, O (-3) → L, O (-3) → L, R (-3) → O → HELLO**
- **S (-3) → P, O (-3) → L, D (-3) → A, Q (-3) → N, U (-3) → R, V (-3) → S → PLANRS**

Decoded Message: **KHOOR SODQUV → HELLO PLANRS**

Key Takeaways

- **Cryptography** protects information via Encryption (**Plaintext → Ciphertext**) and Decryption.
- **Caesar Cipher** shifts alphabet letters by a constant integer key **k**.
- Alphabet wrap-around is governed by **modular arithmetic (mod 26)**.
- Caesar ciphers are easily cracked because there are only **25 possible shift keys**.
- **RSA Public-Key Cryptography** uses two distinct keys (a public key to lock and a private key to unlock) for strong modern security.

Practice Questions for Students

Question 1: Core Terminology

Define the terms **Plaintext**, **Ciphertext**, **Encryption**, and **Decryption** in your own words. How are these four terms related in a cryptographic pipeline?

Question 2: Caesar Cipher Encryption

Encrypt the following plaintext words using a Caesar cipher with a shift key of $k = 4$:

- a) **MATH**
 - b) **SECRET**
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Question 3: Alphabet Wrap-Around (Modular Arithmetic)

When encrypting letters near the end of the alphabet, wrap-around occurs. Determine the ciphertext letter for each of the following shifts:

- a) Encrypt letter **Z** with a shift key of $k = 2$.
 - b) Encrypt letter **Y** with a shift key of $k = 5$.
 - c) Decrypt letter **B** with a shift key of $k = 3$.
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Question 4: Security Analysis & RSA Concepts

- a) Why is a Caesar cipher considered insecure for modern digital communication? How many total shift keys would an attacker need to test to guarantee breaking it?
- b) In RSA public-key cryptography, explain the roles of the **Public Key** and the **Private Key**. Which key is shared publicly, and which key must be kept secret?